6.3 A Polynomial for $\sqrt{2}+\sqrt{3}+\sqrt{5}$

WORKSHEET

The question

STEP II 2023, Question 4

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself To find a polynomial with $\sqrt{2}$ as a root, you square $x = \sqrt{2}$ to get rid of the square root. How might you get rid of two different square roots, one at a time?

STAGE 2 Three square roots

Ask yourself Let $x = \sqrt{2} + \sqrt{3} + \sqrt{5}$, so $x - \sqrt{5} = \sqrt{2} + \sqrt{3}$ is a root of f. Write $f(x - \sqrt{5}) = 0$ in the form $P(x) + \sqrt{5}\,Q(x) = 0$. How do you then eliminate $\sqrt{5}$?

STAGE 3 A cubic and a square root

Ask yourself If $x = a + \sqrt{2}$, then $x - \sqrt{2}$ is a root of $t^3 - 3t + 1$. Substitute, separate the terms with and without $\sqrt{2}$, and square.

STAGE 4 **Two cube roots**

Pause and try Let $x = \sqrt[3]{2} + \sqrt[3]{3}$. Cube both sides. Can you write the result in terms of x and a single cube root?

What's really going on

Fields and their degrees

The set $\mathbb{Q}(\sqrt{2})$ of all numbers $p + q\sqrt{2}$ with p, q rational is closed under addition, subtraction, multiplication and division (by non-zero numbers): it is a *field*. It is a two-dimensional "vector space" over $\mathbb{Q}$, with basis $1, \sqrt{2}$, so we say it has *degree* 2 over $\mathbb{Q}$. Adjoining $\sqrt{3}$ gives $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, with basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ and degree 4; adjoining $\sqrt{5}$ as well gives degree 8 (Figure 6.3.1). The *tower law* says degrees multiply up a tower of fields. The degree of the field generated by a number equals the degree of its *minimal polynomial*, the lowest-degree polynomial with rational coefficients that it satisfies. One can show that the polynomials f and g in this question are minimal, so $\sqrt{2}+\sqrt{3}$ has degree 4 and $\sqrt{2}+\sqrt{3}+\sqrt{5}$ has degree 8: in fact the single number $\sqrt{2}+\sqrt{3}+\sqrt{5}$ generates the whole degree-8 field.

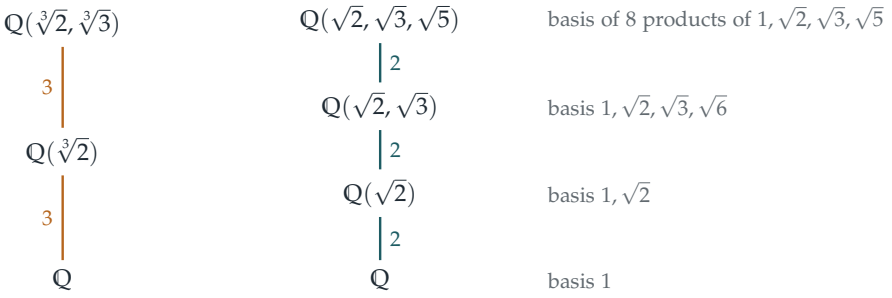


Figure 6.3.1 Towers of fields. Right: each square root doubles the degree, giving $2 \times 2 \times 2 = 8$, the degree of g. Left: each cube root triples it, giving $3 \times 3 = 9$, the degree of k. Degrees multiply up a tower; this *tower law* is the key to the impossibility proofs for ruler-and-compass constructions.

Doubling the cube

Ruler-and-compass constructions can only produce lengths obtained from rational numbers by repeatedly solving linear and quadratic equations. Each such step gives a field extension of degree 1 or 2, so by the tower law every constructible length lies in a field of degree a power of 2, and has degree a power of 2 itself. But $\sqrt[3]{2}$ has minimal polynomial $x^3 - 2$, of degree 3. Since 3 is not a power of 2, $\sqrt[3]{2}$ is not constructible, and the Delian problem of doubling the volume of a cube with ruler and compasses, posed around 430 BC, is impossible. Pierre Wantzel proved this in 1837, together with the impossibility of trisecting a general angle and the classification of constructible polygons (Section 6.2). Two thousand years of attempts were ended by an argument about degrees.

Algebraic numbers form a field

The method of this question shows, in examples, that sums of algebraic numbers are algebraic. In general this is proved with linear algebra: if α and β have degrees m and n, then all the numbers $\alpha^i\beta^j$ (with $i < m$, $j < n$) span a space of dimension at most mn, and the $mn + 1$ powers $1, \alpha + \beta, (\alpha + \beta)^2, \ldots$ must be linearly dependent, which gives a polynomial of degree at most mn. Our degrees $4 = 2 \times 2$, $6 = 3 \times 2$ and $9 = 3 \times 3$ fit this pattern exactly. Numbers that are *not* algebraic are called *transcendental*; e and π are examples (Hermite, 1873, and Lindemann, 1882), and Lindemann's theorem that π is transcendental proved that squaring the circle is impossible too.

Try this next

1 **A primitive element.** Let $y = \sqrt{2} + \sqrt{3}$. Show that $y^3 = 11\sqrt{2} + 9\sqrt{3}$, and deduce that $\sqrt{2} = \frac{1}{2}(y^3 - 9y)$. Hence every number $p + q\sqrt{2} + r\sqrt{3} + s\sqrt{6}$ can be written as a polynomial in y.

2 **Mixed roots.** Find a polynomial of degree 6 with integer coefficients that has $\sqrt{2} + \sqrt[3]{3}$ as a root.

3 **Irreducibility.** Show that $x^4 - 10x^2 + 1$ has no rational roots and cannot be written as a product of two quadratics with integer coefficients. (*Hint:* try $(x^2 + ax + b)(x^2 - ax + c)$.) Deduce that $\sqrt{2} + \sqrt{3}$ has degree 4.

4 **Trisection.** Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$, and deduce that $\cos 20^\circ$ is a root of $8x^3 - 6x - 1 = 0$. Assuming this cubic has no rational roots (check this), explain why a 60° angle cannot be trisected with ruler and compasses.

Further reading

► **Ian Stewart, *Galois Theory*** (CRC Press). Field extensions, the tower law and the classical construction problems, written for undergraduates.

- **Richard Courant & Herbert Robbins, *What Is Mathematics?*** (Oxford University Press). Chapter 3 proves the impossibility of the Greek problems with no prerequisites.
- **Robin Hartshorne, *Geometry: Euclid and Beyond*** (Springer). Ruler-and-compass constructions treated with modern algebra.

The full guided solution is in Section 6.3:
https://beyondstep.keithmcnulty.org/ch06/field-extensions.html